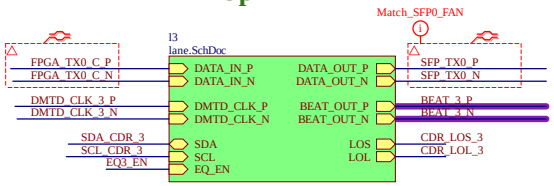
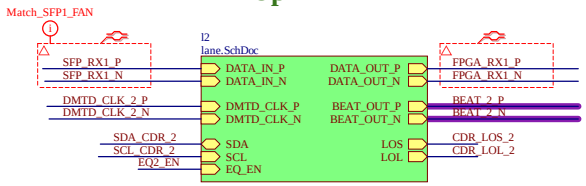
Dedicated to GT 0 (Channel 0) on Quad 0 and SFP 0

UpLink Tx



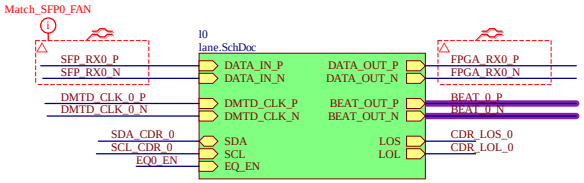
Dedicated to GT 1 (Channel 1) on Quad 0 and SFP 1

UpLink Rx

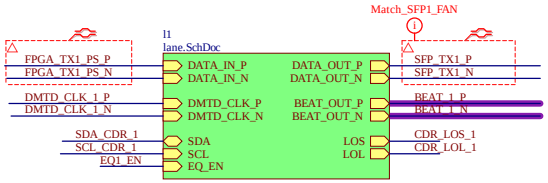


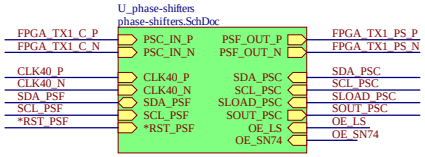
For a new version: 1. add a D-FF to measure the PLL output clock. 2. Remove the phase shifter (does not work on data) and possibly replace it with a working one. 3. Redesign the voltage divider for VREFAC, use a divider per D-FF (see VREFAC note). 4. replace all 0201 and 0102 footprints with larger ones (caps, MOSFETs, and more).

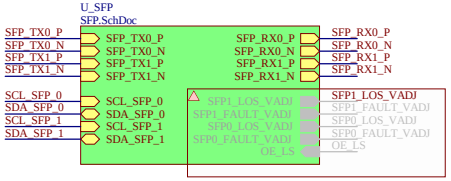
DownLink Rx

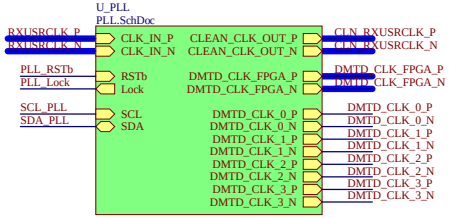


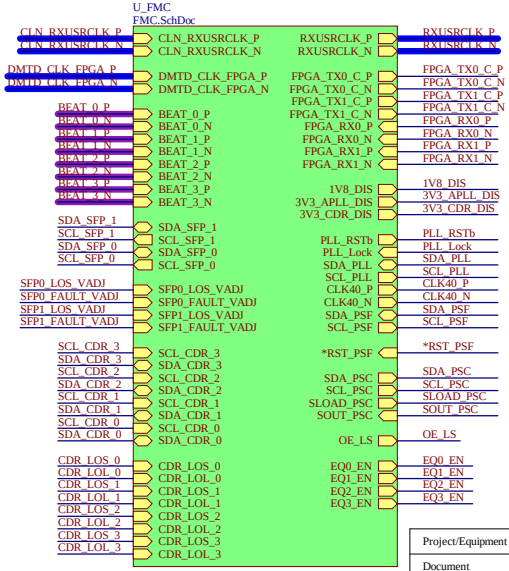
DownLink Tx

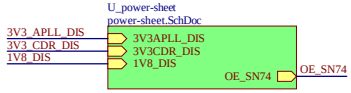












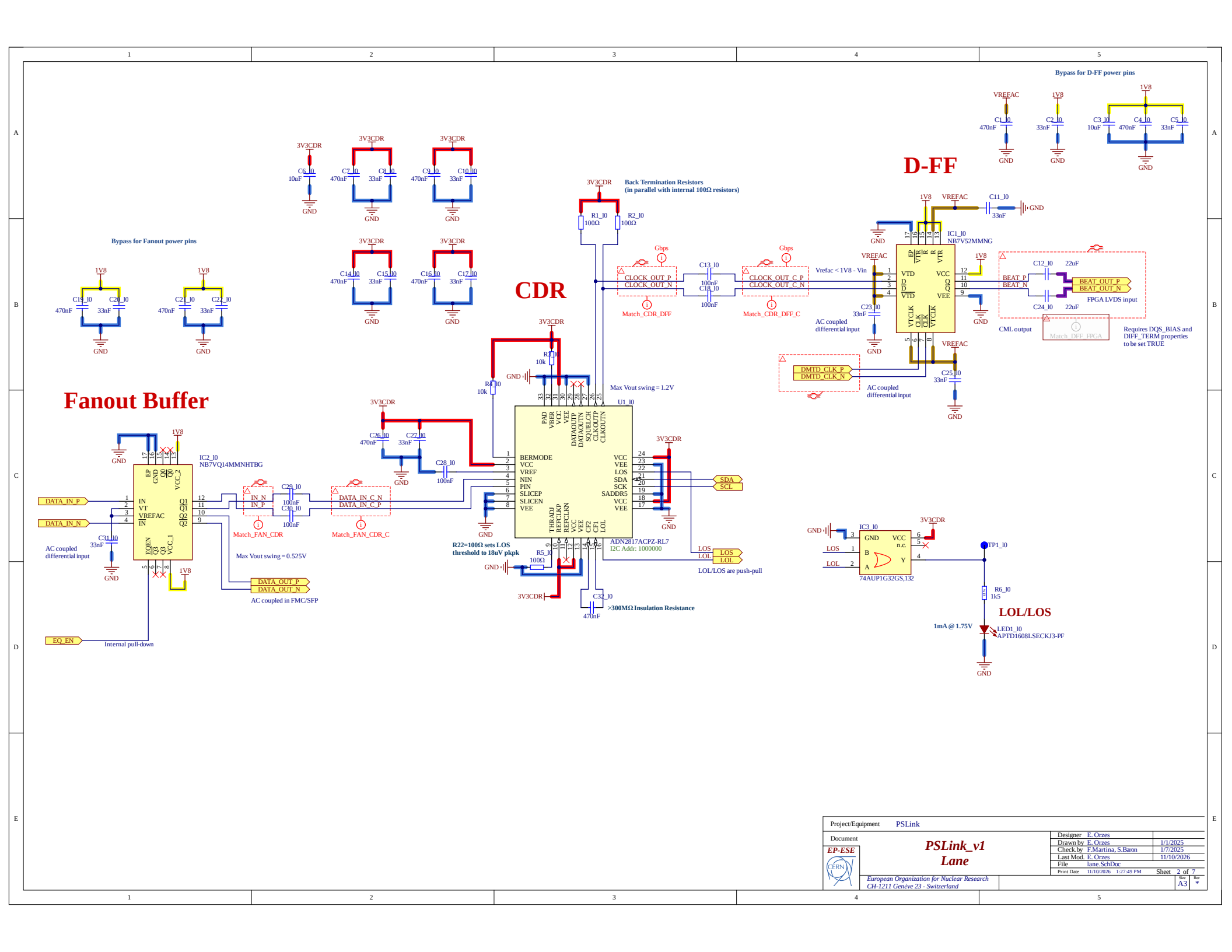
Project Options > Options >

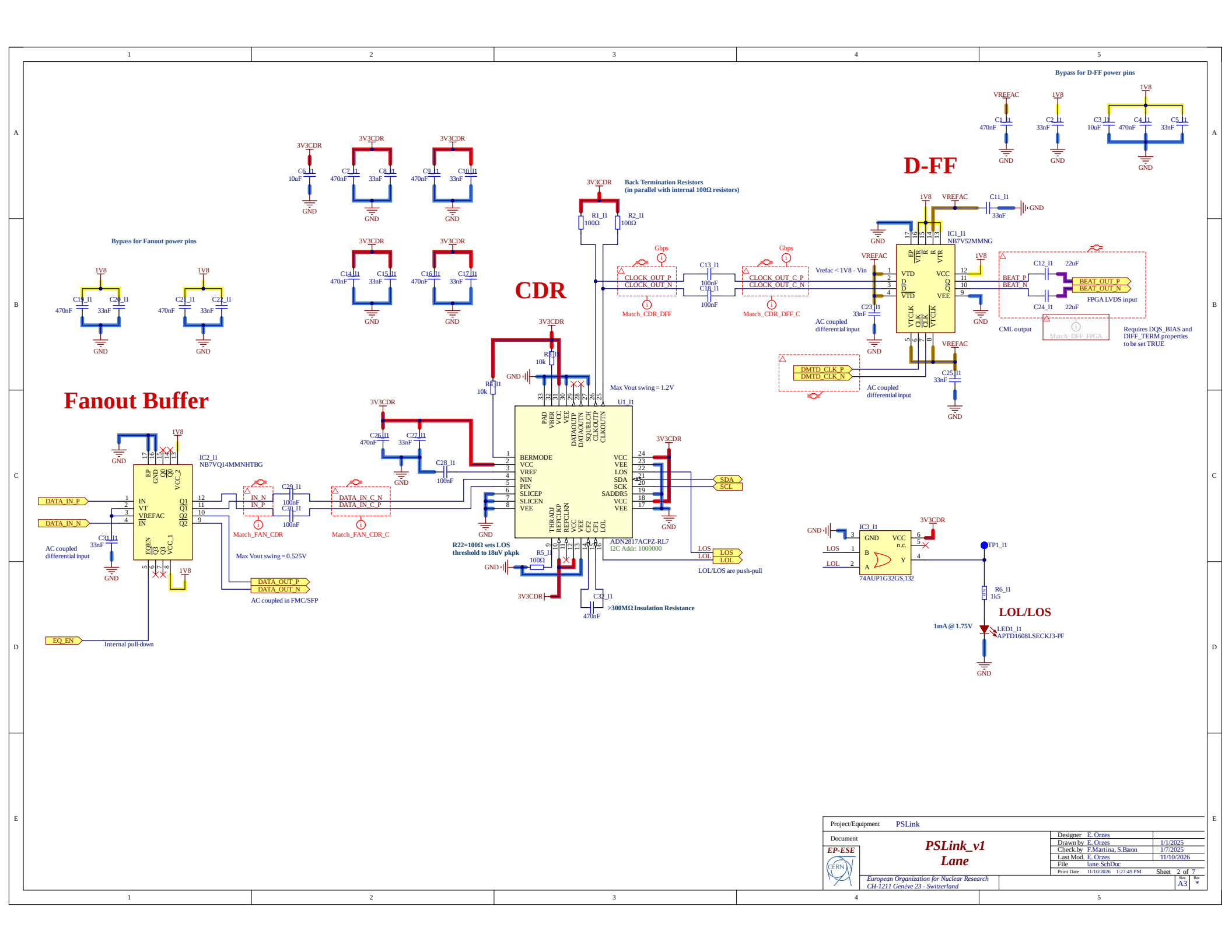
1. Allow ports to name nets (NO)
2. Allow sheet entries to name nets (NO)
3. Append sheet number to local nets (NO)
4. Net identifier scope: Automatic (Hierarchical)
5. Automatic cross reference disabled

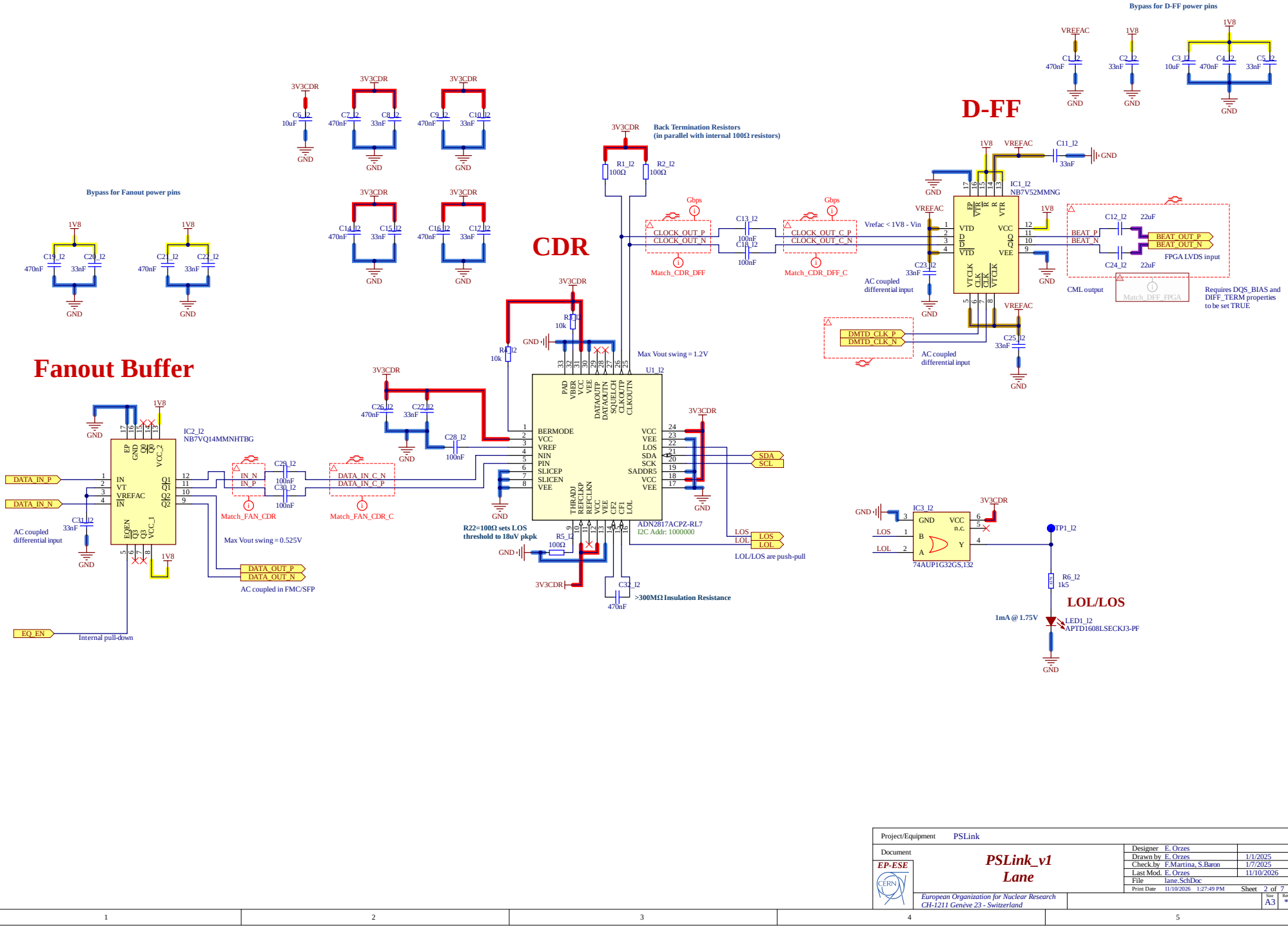
Project/Equipment	PSLink		
Document	**PSLink_v1 Top**	Designer	E. Orzes
		Drawn by	E. Orzes 1/1/2025
		Check.by	F.Martina, S.Baron 1/7/2025
		Last Mod.	E. Orzes 11/10/2026
		File	top.SchDoc
		Print Date	11/10/2026 1:27:49 PM
		Sheet	1 of 7
		Size	A3
		Rev	*

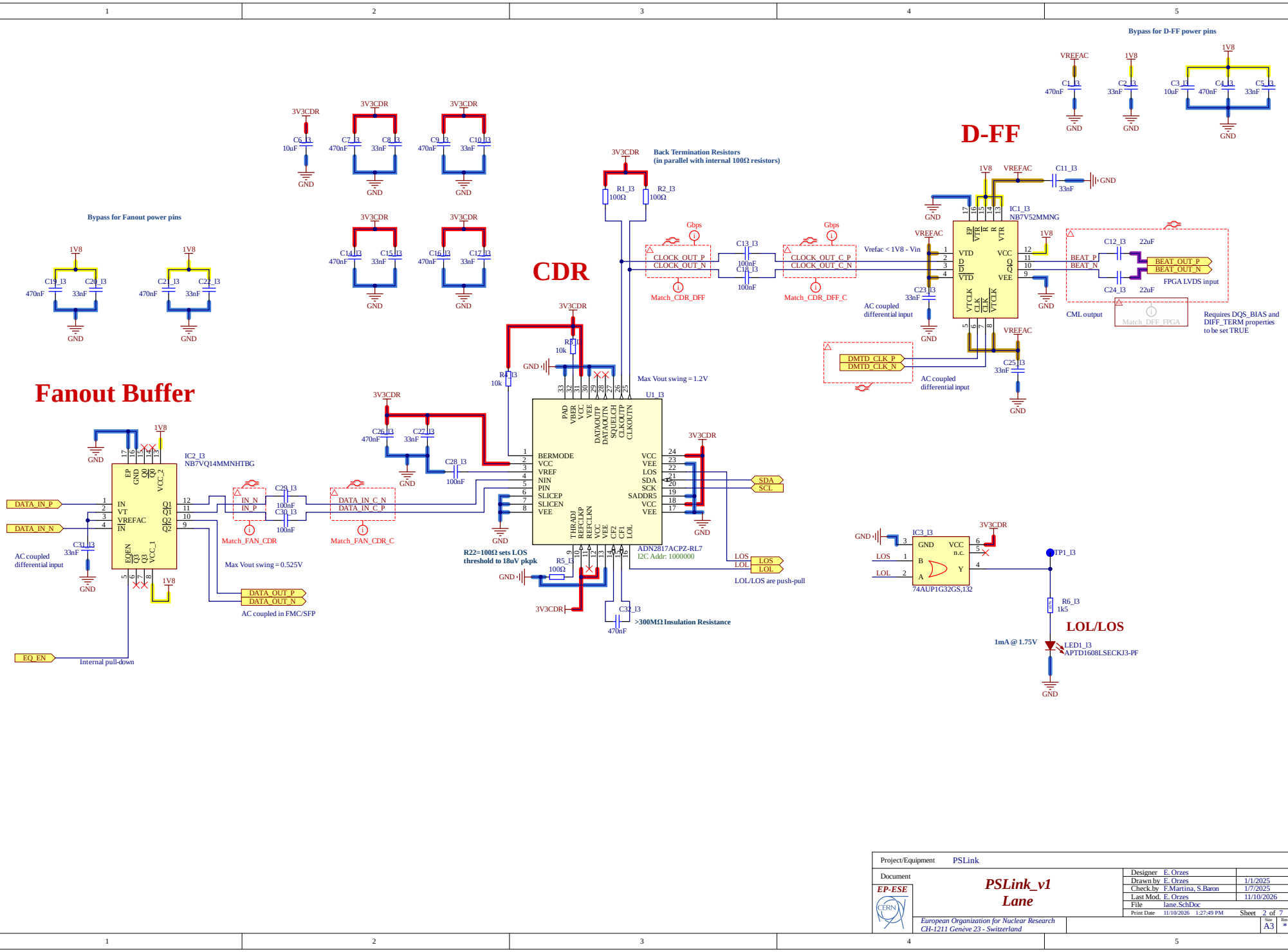
EP-ESE

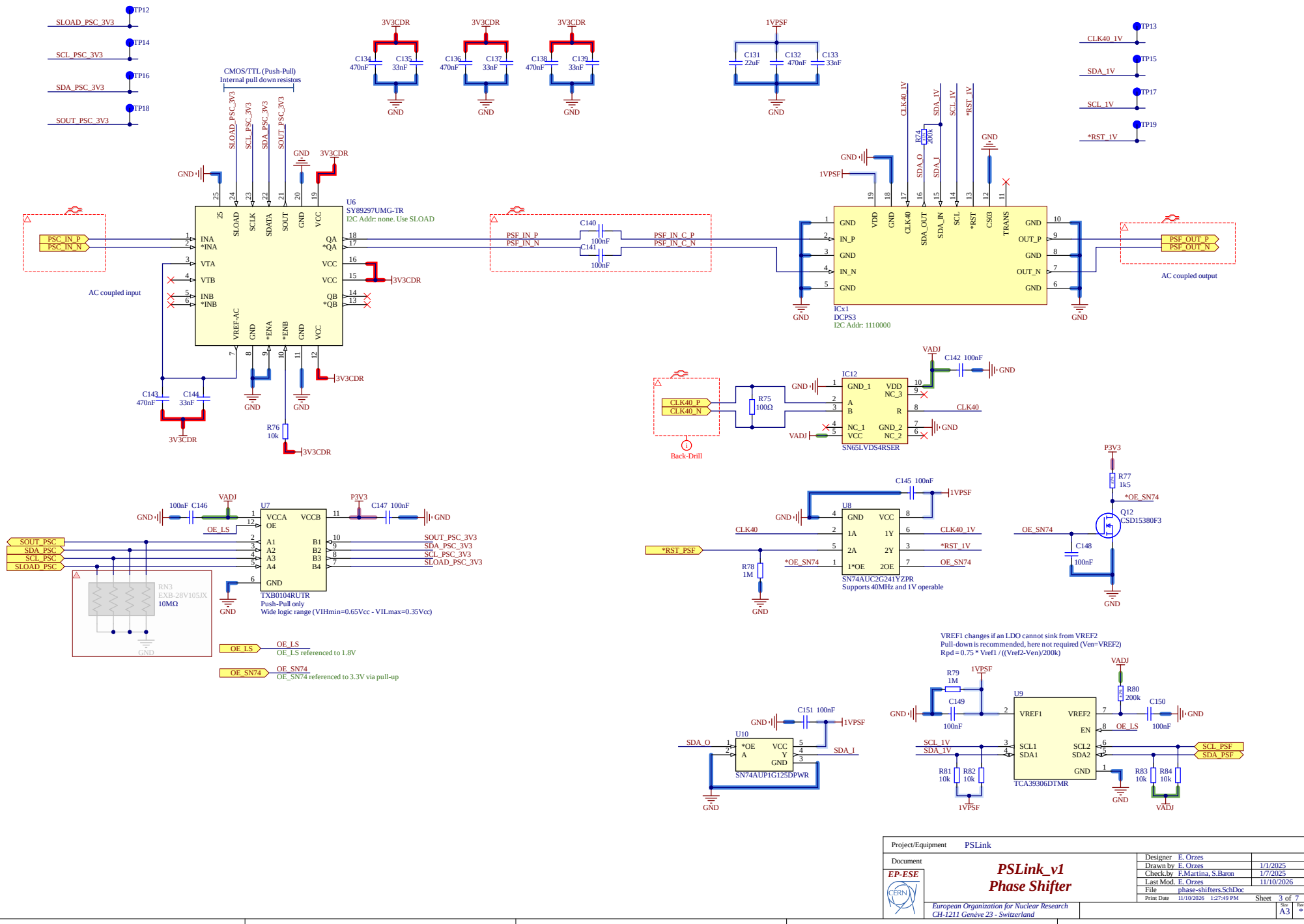
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CH-1211 Genève 23 - Switzerland


Bypass for Fanout power pins
Bypass for D-FF power pins
Fanout Buffer
CDR
D-FF
Back Termination Resistors
(in parallel with internal 100Ω resistors)
Max Vout swing = 1.2V
Max Vout swing = 0.525V
AC coupled differential input
AC coupled in FMC/SFP
Internal pull-down
R22=100Ω sets LOS threshold to 18uV pkpk
LOL/LOS are push-pull
>300MΩ Insulation Resistance
ADN2817ACPZ-RL7
I2C Addr: 1000000
NB7VQ14MMNHTBG
NB7V52MMNG
74AUP1G32GS,132
Vrefac < 1V8 - Vin
FPGA LVDS input
CML output
Requires DQS_BIAS and DIFF_TERM properties to be set TRUE
1mA @ 1.75V
LOL/LOS
Project/Equipment PSLink
Document
PSLink_v1
Lane
Designer E. Orzes
Drawn by E. Orzes 1/1/2025
Check.by F.Martina, S.Baron 1/7/2025
Last Mod. E. Orzes 11/10/2026
File lane.SchDoc
Print Date 11/10/2026 1:27:49 PM
Sheet 2 of 7
Size A3
Rev *
EP-ESE
European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland


Fanout Buffer
CDR
D-FF
Bypass for Fanout power pins
Bypass for D-FF power pins
Back Termination Resistors
(in parallel with internal 100Ω resistors)
IC2_I1
NB7VQ14MMNHTBG
U1_I1
ADN2817ACPZ-RL7
I2C Addr: 1000000
IC1_I1
NB7V52MMNG
IC3_I1
74AUP1G32GS,132
DATA_IN_P
DATA_IN_N
DATA_OUT_P
DATA_OUT_N
DATA_IN_C_P
DATA_IN_C_N
CLOCK_OUT_P
CLOCK_OUT_N
CLOCK_OUT_C_P
CLOCK_OUT_C_N
DMTD_CLK_P
DMTD_CLK_N
BEAT_OUT_P
BEAT_OUT_N
SDA
SCL
LOS
LOL
EQ_EN
Match_FAN_CDR
Match_FAN_CDR_C
Match_CDR_DFF
Match_CDR_DFF_C
Match_DFF_FPGA
AC coupled differential input
AC coupled in FMC/SFP
Internal pull-down
Max Vout swing = 1.2V
Max Vout swing = 0.525V
R22=100Ω sets LOS threshold to 18uV pkpk
LOL/LOS are push-pull
>300MΩ Insulation Resistance
Vrefac < 1V8 - Vin
FPGA LVDS input
CML output
Requires DQS_BIAS and DIFF_TERM properties to be set TRUE
1mA @ 1.75V
LOL/LOS
Project/Equipment PSLink
Document
PSLink_v1
Lane
EP-ESE
European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland
Designer E. Orzes
Drawn by E. Orzes 1/1/2025
Check.by F.Martina, S.Baron 1/7/2025
Last Mod. E. Orzes 11/10/2026
File lane.SchDoc
Print Date 11/10/2026 1:27:49 PM
Sheet 2 of 7
Size A3
Rev *


Bypass for Fanout power pins
Bypass for D-FF power pins
Fanout Buffer
CDR
D-FF
Back Termination Resistors
(in parallel with internal 100Ω resistors)
IC2_I2
NB7VQ14MMNHTBG
U1_I2
ADN2817ACPZ-RL7
I2C Addr: 1000000
IC1_I2
NB7V52MMNG
IC3_I2
74AUP1G32GS,132
Max Vout swing = 1.2V
Max Vout swing = 0.525V
AC coupled differential input
AC coupled in FMC/SFP
Internal pull-down
Vrefac < 1V8 - Vin
FPGA LVDS input
CML output
Requires DQS_BIAS and DIFF_TERM properties to be set TRUE
R22=100Ω sets LOS threshold to 18uV pkpp
>300MΩ Insulation Resistance
LOL/LOS are push-pull
LOL/LOS
1mA @ 1.75V
Match_CDR_DFF
Match_CDR_DFF_C
Match_FAN_CDR
Match_FAN_CDR_C
Match_DFF_FPGA
DATA_IN_P
DATA_IN_N
DATA_OUT_P
DATA_OUT_N
CLOCK_OUT_P
CLOCK_OUT_N
DMTD_CLK_P
DMTD_CLK_N
BEAT_P
BEAT_N
SDA
SCL
LOS
LOL
EQ_EN
Gbps
Project/Equipment PSLink
Document
PSLink_v1
Lane
EP-ESE
European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland
Designer E. Orzes
Drawn by E. Orzes 1/1/2025
Check.by F.Martina, S.Baron 1/7/2025
Last Mod. E. Orzes 11/10/2026
File lane.SchDoc
Print Date 11/10/2026 1:27:49 PM
Sheet 2 of 7
Size A3
Rev *

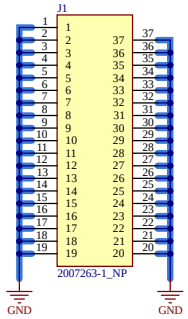

Bypass for Fanout power pins
Bypass for D-FF power pins
Fanout Buffer
CDR
D-FF
Back Termination Resistors
(in parallel with internal 100Ω resistors)
IC2_I3
NB7VQ14MMNHTBG
U1_I3
ADN2817ACPZ-RL7
I2C Addr: 1000000
IC1_I3
NB7V52MMNG
IC3_I3
74AUP1G32GS,132
Max Vout swing = 1.2V
Max Vout swing = 0.525V
AC coupled differential input
AC coupled in FMC/SFP
Internal pull-down
Vrefac < 1V8 - Vin
CML output
FPGA LVDS input
Requires DQS_BIAS and DIFF_TERM properties to be set TRUE
R22=100Ω sets LOS threshold to 18uV pkpp
LOL/LOS are push-pull
>300MΩ Insulation Resistance
1mA @ 1.75V
LOL/LOS
Match_CDR_DFF
Match_CDR_DFF_C
Match_FAN_CDR
Match_FAN_CDR_C
Match_DFF_FPGA
DATA_IN_P
DATA_IN_N
DATA_OUT_P
DATA_OUT_N
CLOCK_OUT_P
CLOCK_OUT_N
DMTD_CLK_P
DMTD_CLK_N
BEAT_P
BEAT_N
SDA
SCL
LOS
LOL
EQ_EN
Project/Equipment PSLink
Document
PSLink_v1
Lane
European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland
Designer E. Orzes
Drawn by E. Orzes 1/1/2025
Check.by F.Martina, S.Baron 1/7/2025
Last Mod. E. Orzes 11/10/2026
File lane.SchDoc
Print Date 11/10/2026 1:27:49 PM
Sheet 2 of 7
Size A3
Rev *

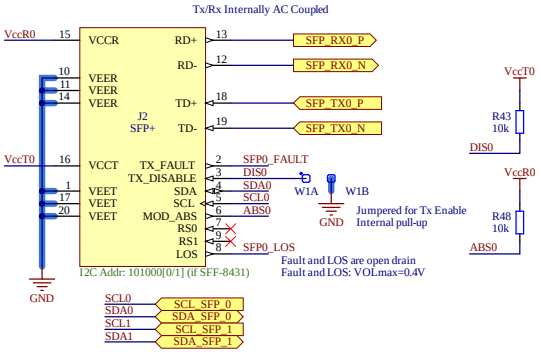


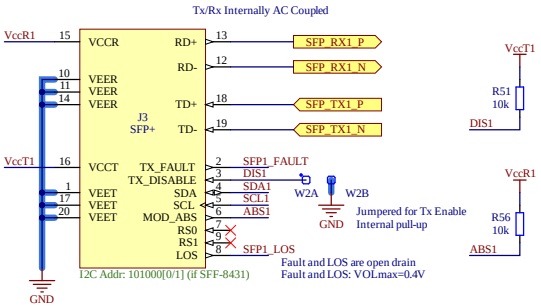
Project/Equipment	PSLink
Document	**PSLink_v1** **Phase Shifter**
Designer	E. Orzes
Drawn by	E. Orzes 1/1/2025
Check.by	F.Martina, S.Baron 1/7/2025
Last Mod.	E. Orzes 11/10/2026
File	phase-shifters.SchDoc
Print Date	11/10/2026 1:27:49 PM
Sheet	3 of 7
Size	A3
Rev	*

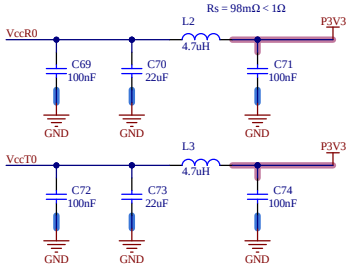
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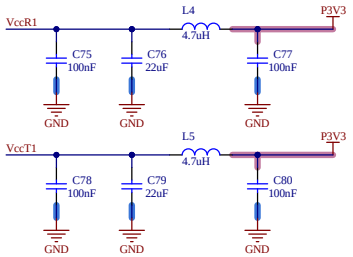
European Organization for Nuclear Research
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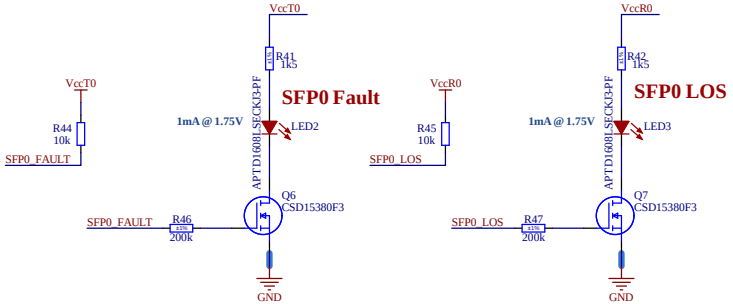


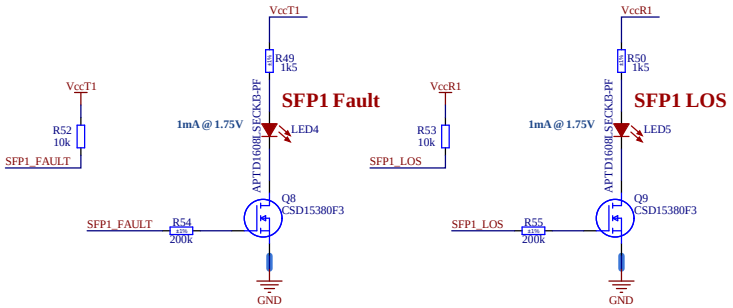


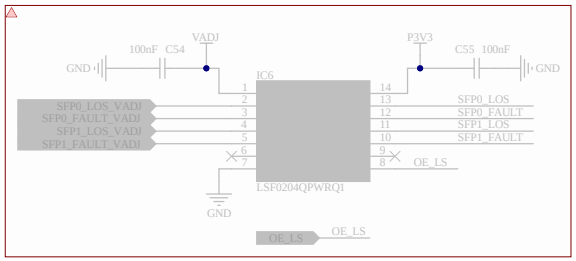




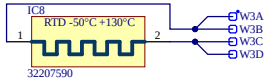








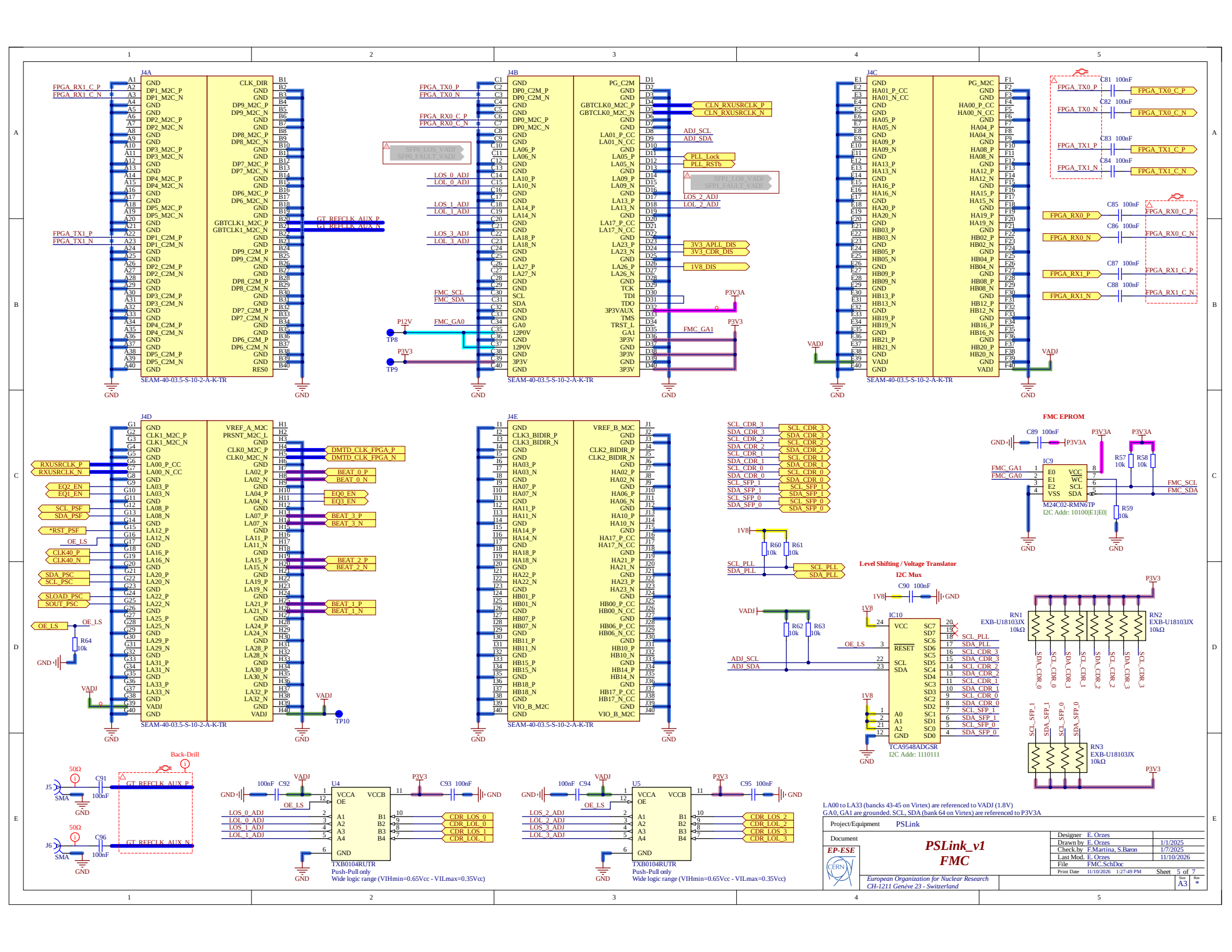
Temperature Sensor



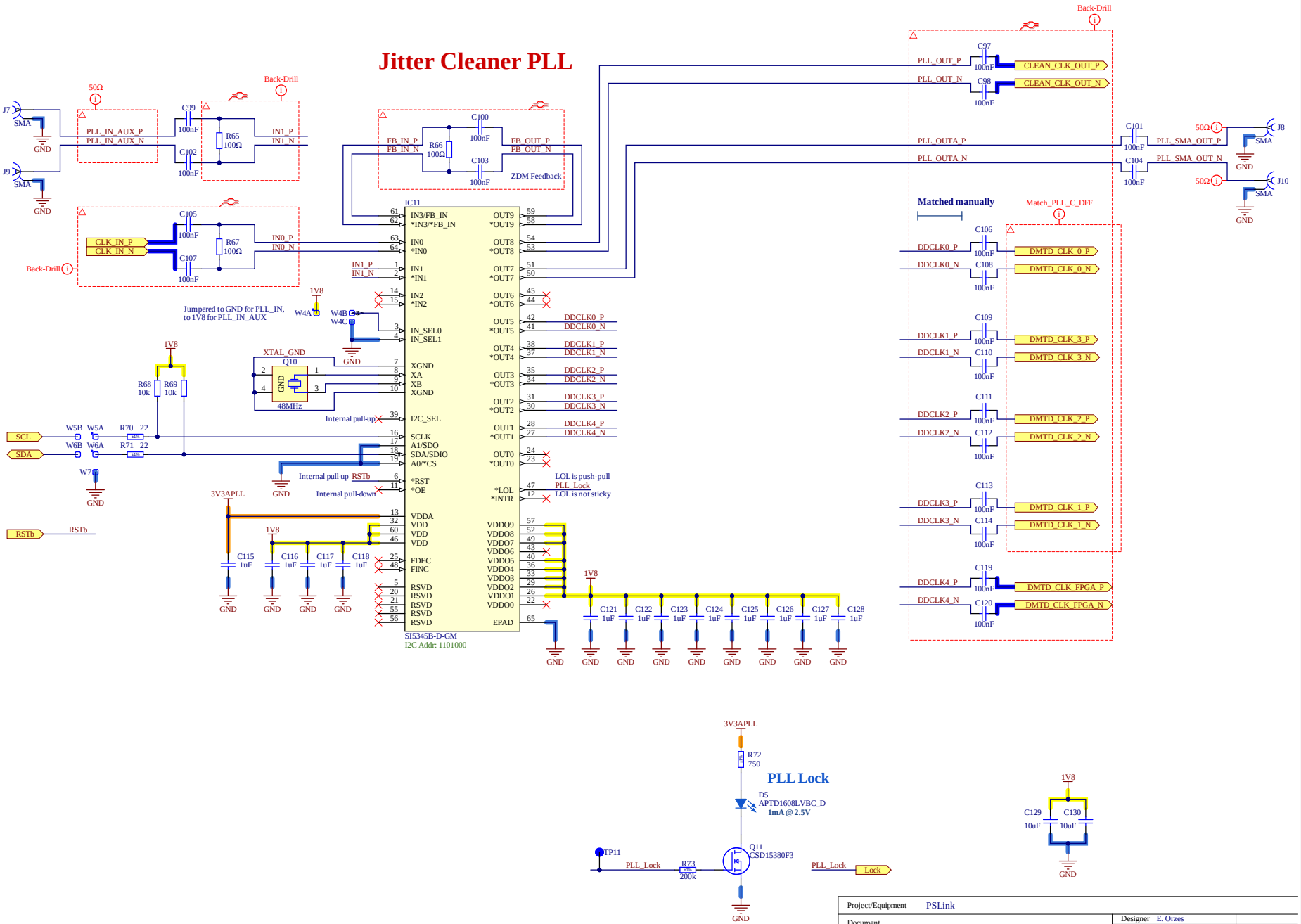
Project/Equipment	PSLink
Document	PSLink_v1 SFP
Designer	E. Orzes
Drawn by	E. Orzes 1/1/2025
Check.by	F.Martina, S.Baron 1/7/2025
Last Mod.	E. Orzes 11/10/2026
File	SFP.SchDoc
Print Date	11/10/2026 1:27:49 PM
Sheet	4 of 7
Size	A3
Rev	*

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CH-1211 Genève 23 - Switzerland


J4A
SEAM-40-03.5-S-10-2-A-K-TR
FPGA_RX1_C_P
FPGA_RX1_C_N
FPGA_TX1_P
FPGA_TX1_N
GT_REFCLK_AUX_P
GT_REFCLK_AUX_N
J4B
FPGA_TX0_P
FPGA_TX0_N
FPGA_RX0_C_P
FPGA_RX0_C_N
SFP0_LOS_VADJ
SFP0_FAULT_VADJ
LOS_0_ADJ
LOL_0_ADJ
LOS_1_ADJ
LOL_1_ADJ
LOS_3_ADJ
LOL_3_ADJ
FMC_SCL
FMC_SDA
FMC_GA0
P12V
P3V3
TP8
TP9
CLN_RXUSRCLK_P
CLN_RXUSRCLK_N
ADJ_SCL
ADJ_SDA
PLL_Lock
PLL_RSTb
SFP1_LOS_VADJ
SFP1_FAULT_VADJ
LOS_2_ADJ
LOL_2_ADJ
3V3_APLL_DIS
3V3_CDR_DIS
1V8_DIS
FMC_GA1
P3V3A
J4C
VADJ
C81 100nF
C82 100nF
C83 100nF
C84 100nF
C85 100nF
C86 100nF
C87 100nF
C88 100nF
J4D
RXUSRCLK_P
RXUSRCLK_N
EQ2_EN
EQ1_EN
SCL_PSF
SDA_PSF
*RST_PSF
OE_LS
CLK40_P
CLK40_N
SDA_PSC
SCL_PSC
SLOAD_PSC
SOUT_PSC
R64
10k
DMTD_CLK_FPGA_P
DMTD_CLK_FPGA_N
BEAT_0_P
BEAT_0_N
EQ0_EN
EQ3_EN
BEAT_3_P
BEAT_3_N
BEAT_2_P
BEAT_2_N
BEAT_1_P
BEAT_1_N
TP10
J4E
SCL_CDR_3
SDA_CDR_3
SCL_CDR_2
SDA_CDR_2
SCL_CDR_1
SDA_CDR_1
SCL_CDR_0
SDA_CDR_0
SCL_SFP_1
SDA_SFP_1
SCL_SFP_0
SDA_SFP_0
1V8
R60
10k
R61
10k
SCL_PLL
SDA_PLL
FMC EPROM
IC9
M24C02-RMN6TP
I2C Addr: 10100|E1|E0|
C89 100nF
R57
10k
R58
10k
R59
10k
Level Shifting / Voltage Translator
I2C Mux
IC10
TCA9548ADGSR
I2C Addr: 1110111
C90 100nF
R62
10k
R63
10k
RN1
EXB-U18103JX
10kΩ
RN2
EXB-U18103JX
10kΩ
RN3
EXB-U18103JX
10kΩ
Back-Drill
50Ω
J5
J6
SMA
C91
C96
100nF
U4
U5
TXB0104RUTR
Push-Pull only
Wide logic range (VIHmin=0.65Vcc - VILmax=0.35Vcc)
C92
C93
C94
C95
CDR_LOS_0
CDR_LOL_0
CDR_LOS_1
CDR_LOL_1
CDR_LOS_2
CDR_LOL_2
CDR_LOS_3
CDR_LOL_3
LA00 to LA33 (bancks 43-45 on Virtex) are referenced to VADJ (1.8V)
GA0, GA1 are grounded. SCL, SDA (bank 64 on Virtex) are referenced to P3V3A
Project/Equipment PSLink
Document
PSLink_v1
FMC
EP-ESE
European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland
Designer E. Orzes
Drawn by E. Orzes 1/1/2025
Check.by F.Martina, S.Baron 1/7/2025
Last Mod. E. Orzes 11/10/2026
File FMC.SchDoc
Print Date 11/10/2026 1:27:49 PM
Sheet 5 of 7
Size A3
Rev *

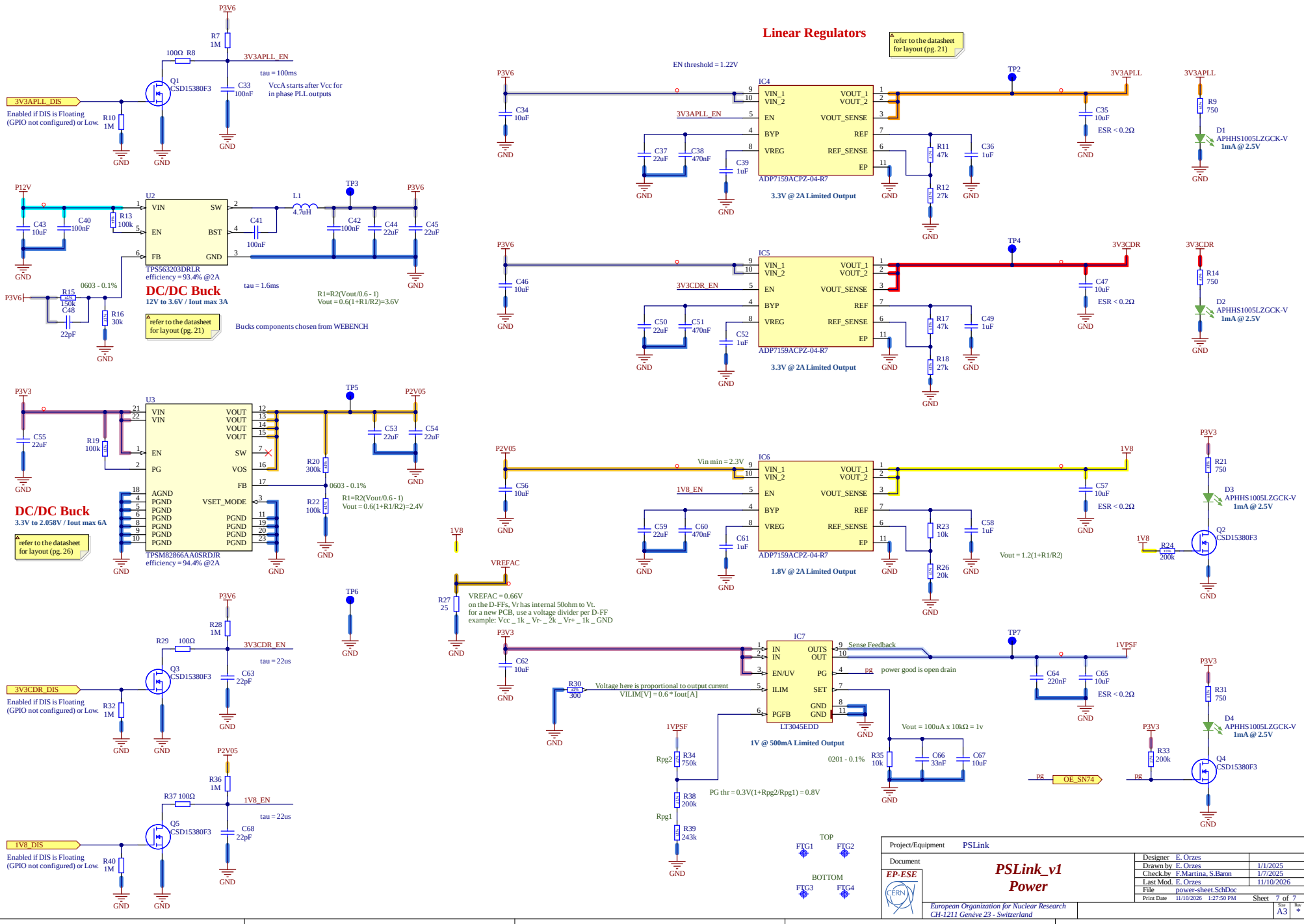
Jitter Cleaner PLL



Project/Equipment	PSLink
Document	PSLink_v1 PLL
Designer	E. Orzes
Drawn by	E. Orzes 1/1/2025
Check.by	F.Martina, S.Baron 1/7/2025
Last Mod.	E. Orzes 11/10/2026
File	PLL.SchDoc
Print Date	11/10/2026 1:27:49 PM
Sheet	6 of 7
Size	A3
Rev	*

EP-ESE

European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland


Linear Regulators
refer to the datasheet for layout (pg. 21)
EN threshold = 1.22V
P3V6
3V3APLL_DIS
Enabled if DIS is Floating (GPIO not configured) or Low.
R10 1M
100Ω R8
R7 1M
3V3APLL_EN
Q1 CSD15380F3
C33 100nF
tau = 100ms
VccA starts after Vcc for in phase PLL outputs
GND
IC4
VIN_1 VIN_2 EN BYP VREG VOUT_1 VOUT_2 VOUT_SENSE REF REF_SENSE EP
ADP7159ACPZ-04-R7
3.3V @ 2A Limited Output
C34 10uF
C37 22uF
C38 470nF
C39 1uF
R11 47k
R12 27k
C36 1uF
TP2
3V3APLL
C35 10uF
ESR < 0.2Ω
R9 750
D1 APHHS1005LZGCK-V 1mA @ 2.5V
P12V
U2
VIN SW EN BST FB GND
TPS563203DRLR
efficiency = 93.4% @2A
C43 10uF
C40 100nF
R13 100k
C41 100nF
L1 4.7uH
TP3
C42 100nF
C44 22uF
C45 22uF
DC/DC Buck
12V to 3.6V / Iout max 3A
tau = 1.6ms
R1=R2(Vout/0.6 - 1)
Vout = 0.6(1+R1/R2)=3.6V
R15 150k
C48 22pF
R16 30k
0603 - 0.1%
refer to the datasheet for layout (pg. 21)
Bucks components chosen from WEBENCH
IC5
3V3CDR_EN
C46 10uF
C50 22uF
C51 470nF
C52 1uF
R17 47k
R18 27k
C49 1uF
TP4
3V3CDR
C47 10uF
R14 750
D2 APHHS1005LZGCK-V 1mA @ 2.5V
P3V3
U3
VIN VOUT EN SW PG VOS FB AGND PGND VSET_MODE
TPSM82866AA0SRDJR
efficiency = 94.4% @2A
C55 22uF
R19 100k
TP5
P2V05
C53 22uF
C54 22uF
R20 300k
R22 100k
0603 - 0.1%
R1=R2(Vout/0.6 - 1)
Vout = 0.6(1+R1/R2)=2.4V
DC/DC Buck
3.3V to 2.058V / Iout max 6A
refer to the datasheet for layout (pg. 26)
IC6
Vin min = 2.3V
1V8_EN
C56 10uF
C59 22uF
C60 470nF
C61 1uF
R23 10k
R26 20k
C58 1uF
1V8
C57 10uF
ESR < 0.2Ω
Vout = 1.2(1+R1/R2)
R21 750
D3 APHHS1005LZGCK-V 1mA @ 2.5V
R24 200k
Q2 CSD15380F3
ADP7159ACPZ-04-R7
1.8V @ 2A Limited Output
1V8
VREFAC
R27 25
VREFAC = 0.66V
on the D-FFs, Vr has internal 50ohm to Vt.
for a new PCB, use a voltage divider per D-FF
example: Vcc _ 1k _ Vr- _ 2k _ Vr+ _ 1k _ GND
TP6
P3V6
R28 1M
R29 100Ω
3V3CDR_EN
tau = 22us
Q3 CSD15380F3
C63 22pF
3V3CDR_DIS
Enabled if DIS is Floating (GPIO not configured) or Low.
R32 1M
P2V05
R36 1M
R37 100Ω
1V8_EN
Q5 CSD15380F3
C68 22pF
1V8_DIS
R40 1M
IC7
IN IN EN/UV ILIM PGFB OUTS OUT PG SET GND GND
LT3045EDD
1V @ 500mA Limited Output
Sense Feedback
power good is open drain
Voltage here is proportional to output current
VILIM[V] = 0.6 * Iout[A]
R30 300
C62 10uF
C64 220nF
C65 10uF
ESR < 0.2Ω
TP7
1VPSF
R34 750k
R38 200k
R39 243k
PG thr = 0.3V(1+Rpg2/Rpg1) = 0.8V
Vout = 100uA x 10kΩ = 1v
R35 10k
C66 33nF
C67 10uF
0201 - 0.1%
pg
OE_SN74
R31 750
D4 APHHS1005LZGCK-V 1mA @ 2.5V
R33 200k
Q4 CSD15380F3
TOP
BOTTOM
FTG1 FTG2 FTG3 FTG4
Project/Equipment PSLink
Document
PSLink_v1
Power
EP-ESE
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